

Question 1.*(Start a new page.)*

- a) Find, correct to one decimal place, the value of : $\frac{1.09^2 + 0.89}{1.09^2 - 0.89}$.
- b) Express $\frac{2}{3}$ of $\frac{1}{4}$ as an exact percentage.
- c) Simplify $a - 3(a - 2)$.
- d) Find the coordinates of the vertex of the parabola
 $y = 2x^2 - 20x + 53$.
- e) Solve $x^2 - 3x - 28 = 0$.
- f) Evaluate $\log_5 45 + \log_5 40 - \log_5 72$.

Question 2.*(Start a new page.)*

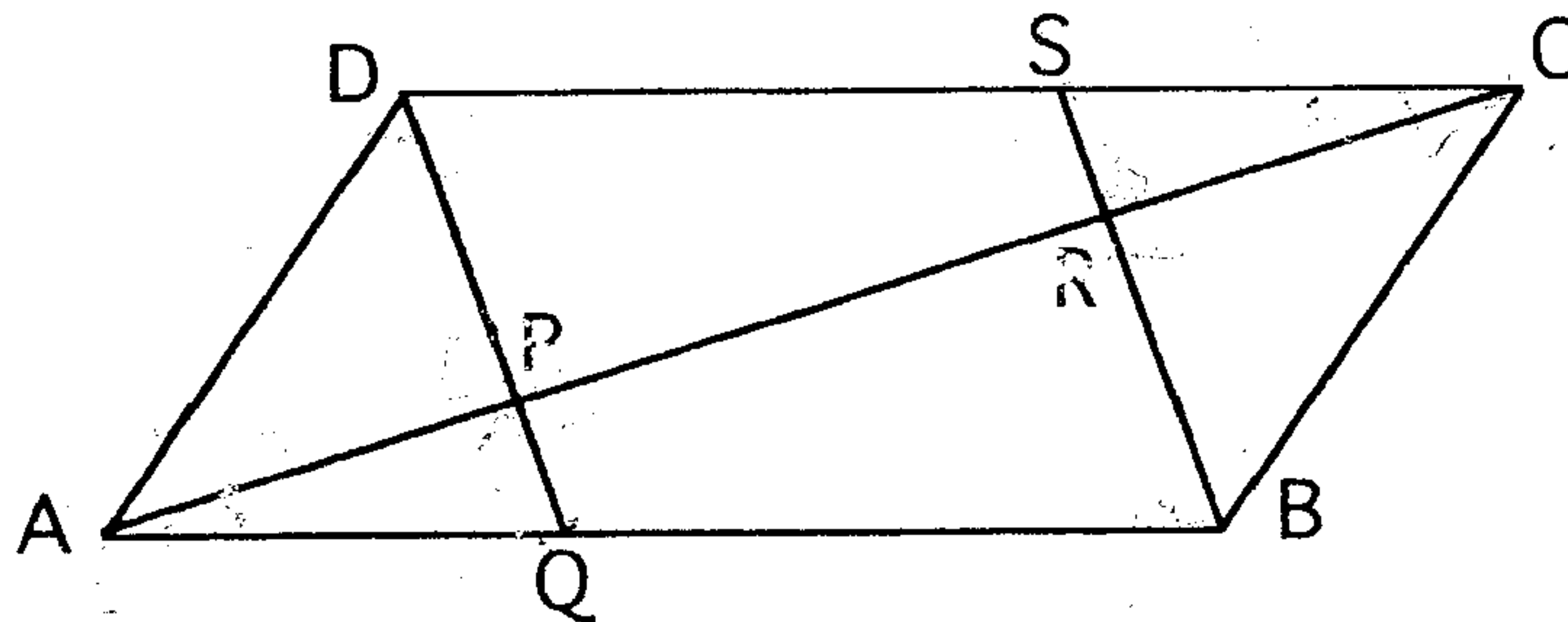
- a) Two lines $l: x + y = 12$ and $m: 2x - y = -3$ intersect at the point M.
- (i) Find the coordinates of M.
- (ii) Prove that the point A(6,6) lies on the line l .
- (iii) The line through A parallel to the y-axis intersects the line m at B. Find the coordinates of B.
- (iv) If lines l and m intersect the y-axis at C(0,12) and D(0,3) respectively, show that the quadrilateral ABCD is a parallelogram. **Give reasons.**
- (v) Hence find the area of the triangle AMD.
- b) Find the limiting sum of the geometric series $\frac{1}{3} + \frac{1}{6} + \frac{1}{12} + \frac{1}{24} + \dots$

Question 3.*(Start a new page.)*

- a) Differentiate :
 (i) $x^2 - \frac{3}{7}$.
 (ii) $\frac{\sin x}{x}$.
 (iii) $(3 - 4x)^5$.
- b) Find the equation of the tangent to the curve $y = x \ln x$ at the point $(1, 0)$.
- c) Find $\int \sec^2 3x \, dx$.
- d) Evaluate $\int_1^3 \frac{2x}{x^2 + 3} \, dx$

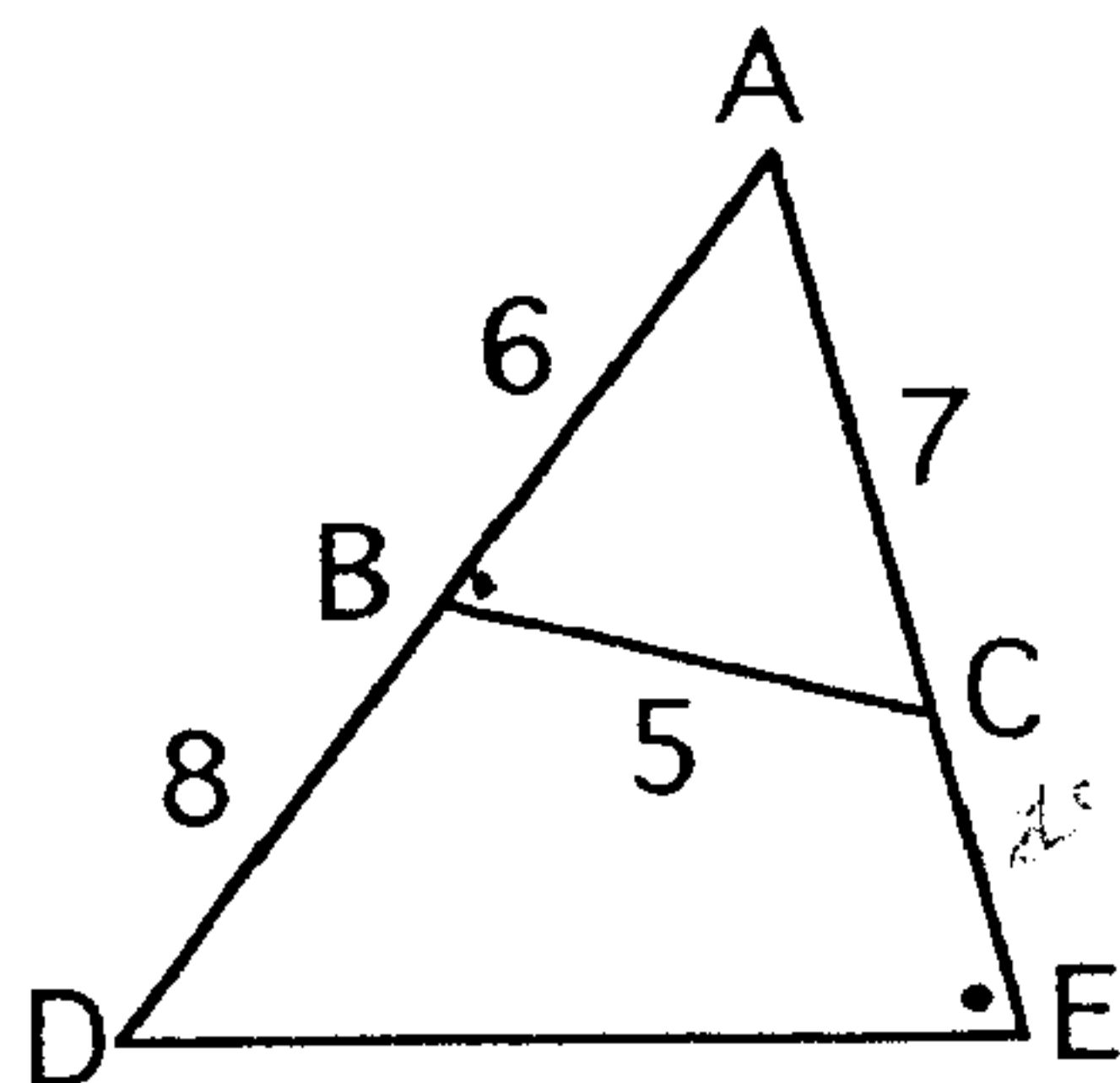
Question 4.*(Start a new page.)*

- a) In the parallelogram ABCD, the straight lines DPQ and BRS are perpendicular to the diagonal AC.



Copy the figure on your answer sheet and prove, giving reasons :

- (i) $DQ \parallel SB$.
 (ii) Quadrilateral QBSD is a parallelogram.
 (iii) $SC = AQ$.
- b) In the figure above, if the area of the triangle AQD is 20 cm^2 and $AQ : QB = 2 : 3$ find the area of the parallelogram ABCD.
- c) In the figure below, not drawn to scale, $\angle ABC = \angle CED$, $AB = 6 \text{ cm}$, $BD = 8 \text{ cm}$, $AC = 7 \text{ cm}$ and $BC = 5 \text{ cm}$.



- (i) Prove that the triangles ABC and ADE are similar.
 (ii) Find the length of CE. Give reasons.

Question 5.(Start a new page.)

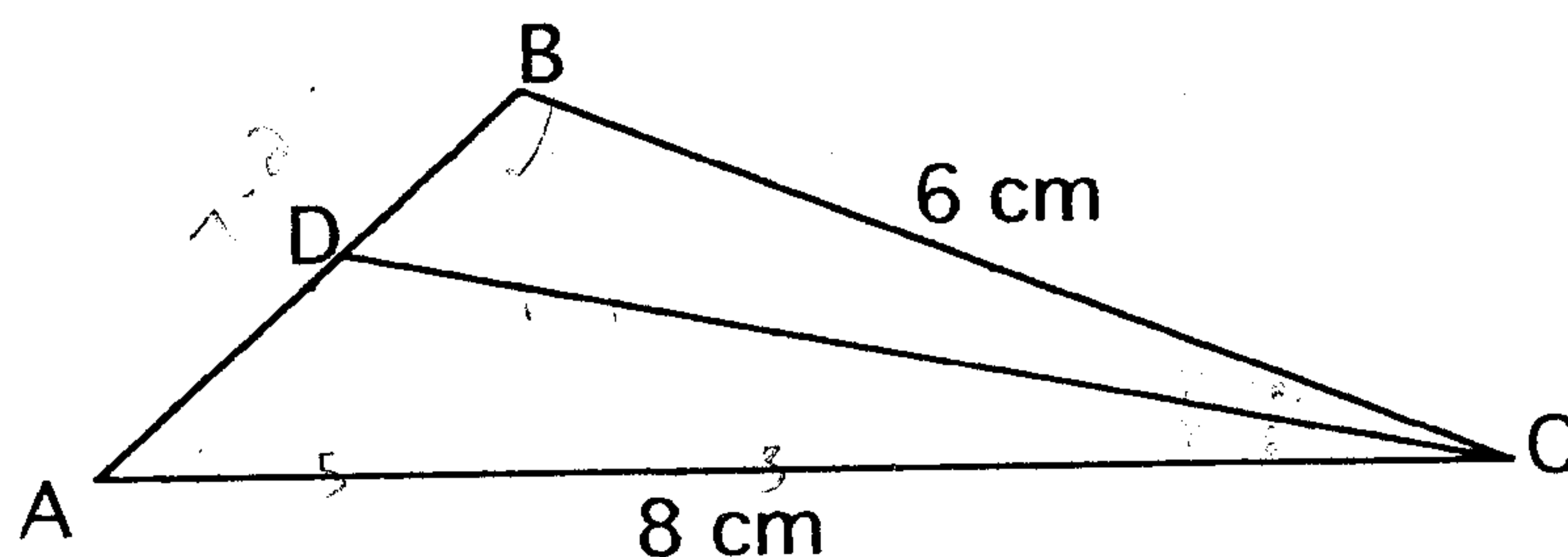
- a) Two 6 sided, unbiased dice, one red and the other green, are tossed. What is the probability that :
- (i) The red one is an odd number ?
 - (ii) Their sum is 8 ?
 - (iii) Their sum is 8 given that the red one shows an odd number ?
- b) A particle initially at ~~rest~~^{origin}, moves in a straight line. The displacement x metres from the origin at time t seconds is given by $x = t^3 - 6t^2 + 9t$, $t \geq 0$.
- (i) Find the velocity of the particle at time t seconds.
 - (ii) Hence prove that the particle changes direction twice in the first 4 seconds, and find the positions of the particle at those times.
 - (iii) Sketch the displacement-time graph showing all information obtained so far, for $t \leq 4$.
 - (iv) Calculate the average speed over the first 3 seconds.

Question 6.(Start a new page.)

- a) The number of bacteria, N , in a culture increased from 600 to 1800 in two hours. Assuming that the rate of increase is directly proportional to the number of the bacteria present, $\frac{dN}{dt} = kN$,
- (i) Show that $N = Ae^{kt}$, where A is a constant, satisfies the equation $\frac{dN}{dt} = kN$.
 - (ii) Find the constant A and then the constant k .
 - (iii) How many hours does it take for the number of bacteria to increase from 1800 to 18000 ?
- b) (i) Sketch the graph of $y = \sqrt{4 - x^2}$.
- (ii) Hence, find $\int_{-2}^2 \sqrt{4 - x^2} dx$.

Question 7.*(Start a new page.)*

- a) A curve $y = f(x)$ has derivative $f'(x) = kx^2 + 2$ and a stationary point at $(-1, 3)$.
- (i) Find k .
- (ii) Find the function.
- b) In $\triangle ABC$, not drawn to scale, CD bisects the angle BCA and D lies on side AB . $\angle C = 60^\circ$, $AC = 8$ cm and $BC = 6$ cm.



- (i) Calculate the length of AB , giving your answer correct to one decimal place.
- (ii) Calculate the exact area of $\triangle ABC$.
- (iii) Hence, calculate the exact length of CD .

Question 8.*(Start a new page.)*

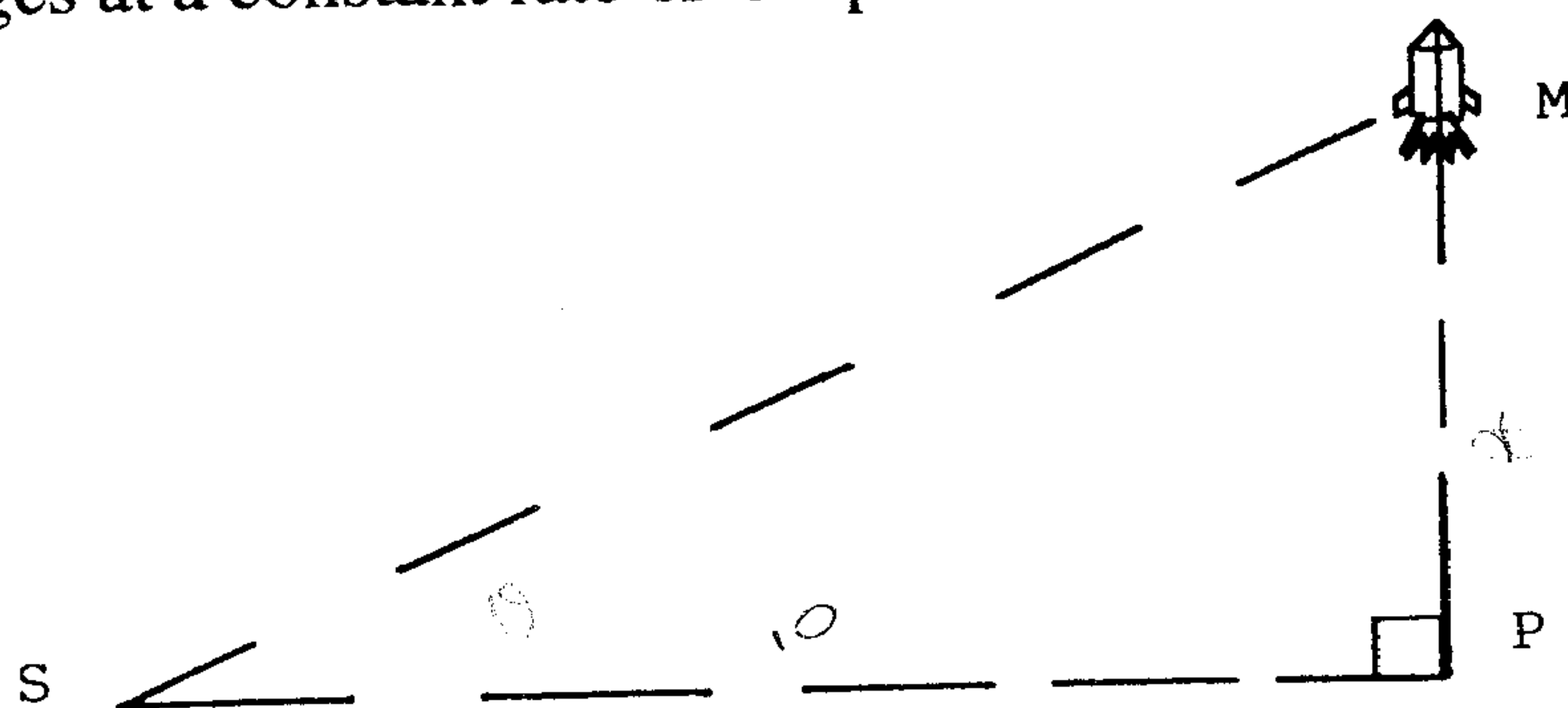
- a) Calculate the area bounded by $y = x^2 - 5x + 6$ and $y = 2x$.
- b) Every year, starting on Peter's first birthday, his grandparents gave him \$100 which was deposited in the same bank account at the rate of 8% per annum. On his 21st birthday, instead of the \$100, they gave him a lump sum of \$5,000, deposited in the same account. After this they stopped giving Peter any more money.
- (i) How much money did Peter have in this account before his 21st birthday?
- (ii) Peter left all this money in the bank, at the usual rate of 8%, until his 28th birthday when he decided to use it for a new car. How much money did he collect from this account?

Question 9. (Start a new page.)

- a) The area in the first quadrant, under the curve $y = xe^x$, the line $x = 2$ and the x-axis, is rotated about the x-axis.
- Write down a definite integral which gives the volume of the generated solid.
 - Calculate an estimate of the volume of this solid of revolution using Simpson's rule with three function values.
- b) Towns A, B and C are located 6 km west, 6 km east and 10 km south, respectively, of a point D. A road is to run north from C to a point P and from P a branch road is to run to A and another branch road to B. (Assume all roads to be straight lines.)
- Show that, the total length L of road $PA + PB + PC$ is given by $L = x + 2\sqrt{x^2 - 20x + 136}$, where x is the length of the road from C to P.
 - Show that the minimum total length for the three roads, $PA + PB + PC$, is $(10 + 6\sqrt{3})$ km.

Question 10. (Start a new page.)

- a) Given the quadratic equation $x^2 - 3x + k = 0$,
- Find the value of k for which the equation has two equal roots.
 - Find the value of k for which one of the roots exceeds the other root by two.
- b) A missile, M, is fired vertically from a point P which is 10 km from a tracking station S at the same elevation as P, as shown below. For the first 20 seconds of flight its angle of elevation θ changes at a constant rate of 2° per second.



- Find θ as a function of t (in the first 20 seconds).
- Hence express the distance x travelled by the missile, as a function of t .
- Find the velocity of the missile when the angle of elevation is 30° , giving your answer in km/hr.

Q1 a) 7.0

b) $16\frac{2}{3}\%$

c) $-2a+6$

d) (5,3)

e) $x=7$ or -4

f) 2

Q2 a) i) M(3,9)

ii) L.H.S. = $6+6=12$ = R.H.S.

iii) B(6,15)

iv) ABCD is a parm.
(one pair of opposite sides equal and parallel)

v) $13\frac{1}{3}$ units²

b) $S_{\infty} = \frac{2}{3}$

Q3 a) i) $2x$

ii) $\frac{x \cos x - \sin x}{x^2}$

iii) $-20(3-4x)^4$

b) $m=1, y=x-1$

c) $\frac{1}{3} \tan 3x + c$

d) $\ln 3$

Q4. a) i) DA || SB (one pair of alternate angles are equal).

ii) ABCD is a parm (2 pairs of opposite sides parallel).

iii) SC = AQ

b) 100 cm^2

c) is equiangular test

ii) CE = 5cm

Q5. i) $\frac{1}{2}$

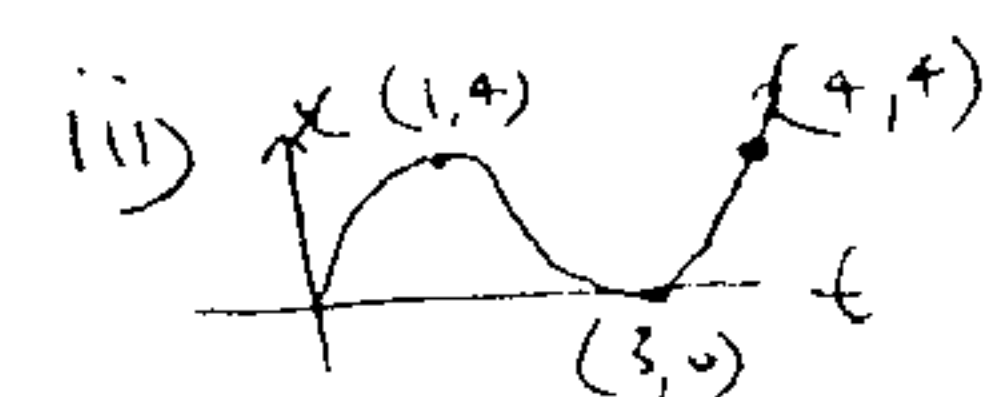
ii) $\frac{5}{36}$

iii) $\frac{2}{18} = \frac{1}{9}$

b) $x = t^3 - 6t^2 + 9t$
 $= t(t-3)^2$

ii) $v = 3t^2 - 12t + 9$

ii) $t=1, t=3$

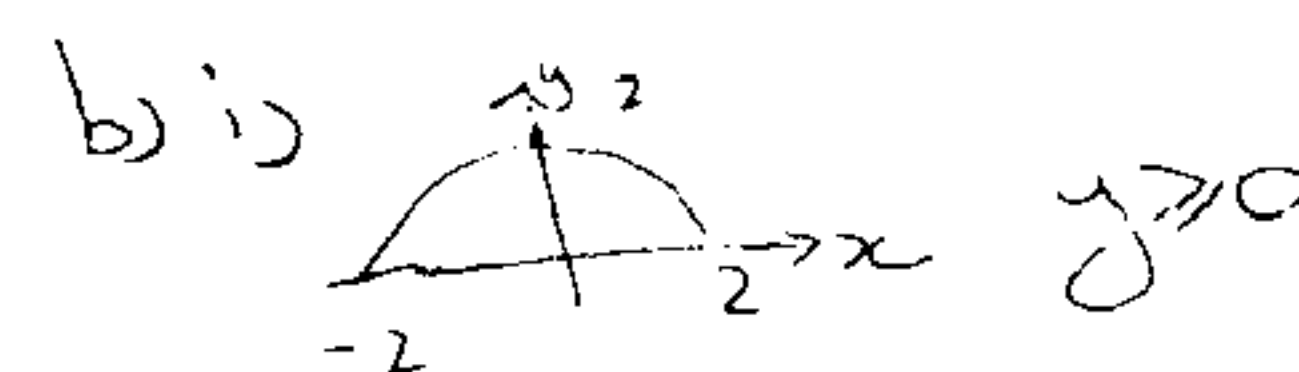


iv) $\frac{4+t}{3} = \frac{8}{3} \text{ m/s}$

Q6. a) i) $\frac{dN}{dt} = AKe^{kt}$
 $= kN$

ii) $A=600, k=\frac{\ln 3}{2}$

iii) $t = 6.2 \text{ hours} - 2$
 $= 4 \text{ hrs } 10 \text{ mins}$



ii) $2\pi \text{ units}^2$

Q7. a) i) $k=-2$

ii) $c = 4\frac{1}{3}$

$f(x) = -\frac{2}{3}x^3 + 2x + 4\frac{1}{3}$

b) i) AB = 7.2cm

ii) Area = $12\sqrt{3} \text{ cm}^2$

iii) DC = $\frac{24\sqrt{3}}{7}$

Q8. i) $x=1, x=6$

$A = 20\frac{5}{6} \text{ units}^2$

b) ± 4.942

ii) $\pm 9.942.29$

worth = $\$17039$

Q9 a) i) $V = \int_0^2 \pi x^2 e^{2x} dx$

ii) 236 units^3

b) i) $PA = x + 2\sqrt{x^2 - 20x + 136}$

ii) min. TP. at $x = 10 - 2\sqrt{3}$

abs. min = $10 - 2\sqrt{3} + 4\sqrt{3} + 4\sqrt{3}$

Q10. a) i) $k = \frac{9}{4}$

ii) $P = \frac{1}{2}(\frac{1}{2} + 2)$
 $= \frac{5}{4}$

b) i) $\theta = (2t)^\circ$

ii) $x = 10 \tan(2t)$
 $x = 10 \tan(\frac{\pi}{90}t)$

iii) $V = 10 \times \frac{\pi}{90} \times \sec^2(\frac{\pi}{90}t)$

$= \frac{4\pi}{27} \text{ km/s}$

$= \frac{1600\pi}{3} \text{ km/h}$